

F78PB Probability and Statistics B

Confidence Intervals

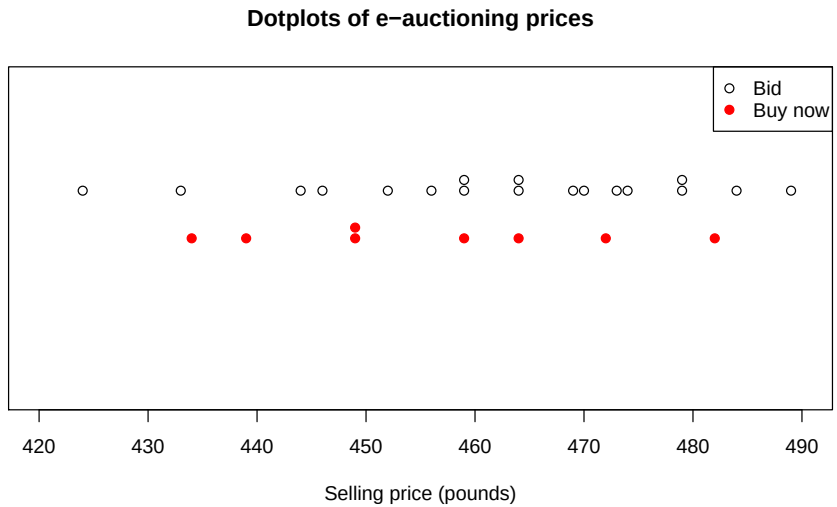
Example

Potential buyers on an Internet auction site can bid for an item over a period of time; they are also offered a *buy-it-now* option at a fixed price. We want to investigate whether selling prices tend to be lower or higher under bidding or the *buy-it-now* option.
The data below are selling prices (in £) for a certain item over one week:
'Buy-it-now': `459 449 449 464 482 472 434 439`
Bid: `444 473 489 479 433 464 452 479 456 470 459 474 464 469 484 446 459 424`

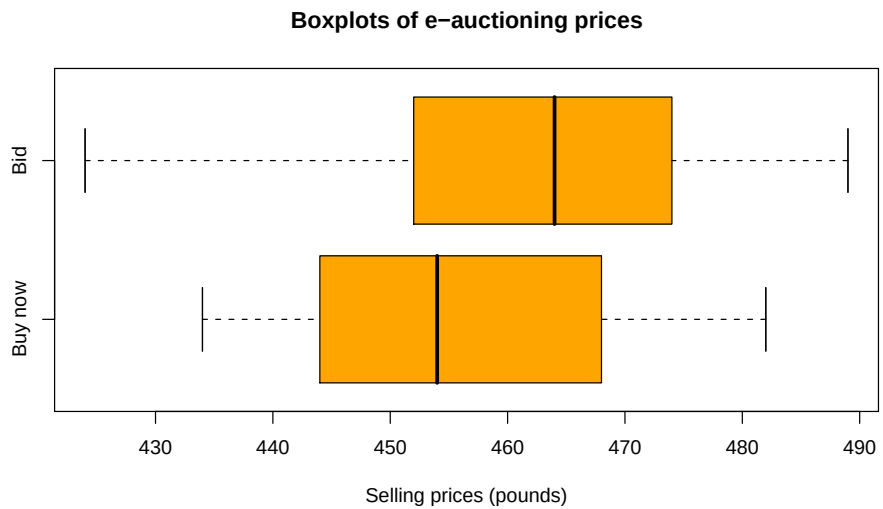
a. Draw a suitable plot for comparing the two data sets.
b. Find a 90% CI for the true difference in the mean price under these two options. Based on this interval, would you support that selling prices tend to be different under the two options?
c. What are the assumptions that the above CI is based on?
d. Check these assumptions using the plot in (a).

Solution

(a) We can use dotplots on the same scale



or boxplots



(b), (c), (d) in class ...

Comparison of variances

Since σ_X^2 and σ_Y^2 are measures of spread (or scale) it is natural to consider the *ratio* σ_X^2/σ_Y^2 when comparing variances.

S_1^2 is a 'good' estimator for σ_1^2 and S_2^2 is a 'good' estimator for σ_2^2, so the ratio S_1^2/S_2^2 is a reasonable estimator for σ_1^2/σ_2^2.

To develop a *confidence interval* for the ratio σ_1^2/σ_2^2, we need to *assume that both populations are normally distributed* and use the F-distribution.

Recall that if $U \sim \chi^2_{n_1}$ and $V \sim \chi^2_{n_2}$ are independent random variables, then $F = \frac{U/n_1}{V/n_2}$ has an F_{n_1,n_2} distribution.

Apply this using:

$$\frac{(n_1-1)S_1^2}{\sigma_1^2} \sim \chi^2_{n_1-1} \quad \text{and} \quad \frac{(n_2-1)S_2^2}{\sigma_2^2} \sim \chi^2_{n_2-1}.$$

So

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} = \frac{S_1^2}{S_2^2}\frac{\sigma_2^2}{\sigma_1^2} \sim F_{n_1-1,n_2-1}.$$

This provides a *pivotal quantity* for the CI for the ratio σ_1^2/σ_2^2.

Comparison of Variances

Goal: Compare population variances using their ratio:

$$\frac{\sigma_X^2}{\sigma_Y^2}.$$

Since S_X^2 and S_Y^2 are 'good' estimators of σ_X^2 and σ_Y^2, the ratio

$$\frac{S_X^2}{S_Y^2}$$

is a reasonable estimator for $\frac{\sigma_X^2}{\sigma_Y^2}$.

To construct a confidence interval for $\frac{\sigma_X^2}{\sigma_Y^2}$, assume:

Both populations are normal.

Recall: If $U \sim \chi^2_{n_1}$ and $V \sim \chi^2_{n_2}$ are independent, then

$$F = \frac{U/n_1}{V/n_2} \sim F_{n_1,n_2}.$$

Apply this using:

$$\frac{(n_X - 1)S_X^2}{\sigma_X^2} \sim \chi^2_{n_X-1}, \qquad \frac{(n_Y - 1)S_Y^2}{\sigma_Y^2} \sim \chi^2_{n_Y-1}.$$

Thus:

$$\frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} = \frac{S_X^2}{S_Y^2} \cdot \frac{\sigma_Y^2}{\sigma_X^2} \sim F_{n_X-1,n_Y-1}.$$

This provides a **pivotal quantity** for constructing a confidence interval for $\frac{\sigma_X^2}{\sigma_Y^2}$.

For an (equal-tailed) $100(1-\alpha)\%$ CI for the ratio σ_X^2/σ_Y^2, we use

$$\mathbb{P}\left(F_{n_X-1,n_Y-1,1-\alpha/2} < \frac{S_X^2}{S_Y^2}\frac{\sigma_Y^2}{\sigma_X^2} < F_{n_X-1,n_Y-1,\alpha/2}\right) = 1-\alpha$$

where $F_{n_1-1,n_2-1,p}$ are percentage points for the appropriate F distribution (use Tables).

Note: Values for $F_{n_1-1,n_2-1,0.975}$ are not in the tables.

To find this value you must use the fact that (in general) for any n and m, $F_{n,m,1-p} = \frac{1}{F_{m,n,p}}$.

We can now show that the $100(1-\alpha)\%$ (equal-tailed) CI for σ_X^2/σ_Y^2 is given by

$$\left(\frac{S_X^2/S_Y^2}{F_{n_X-1,n_Y-1,\alpha/2}}, \frac{S_X^2/S_Y^2}{F_{n_X-1,n_Y-1,1-\alpha/2}}\right).$$

Auctioning Example cont.

For the auctioning data check the assumption of equal population variances between the prices under the two selling options using a suitable 95% CI.

Solution ...

Paired data

The analyses given above depend in a crucial way on the assumption that the 2 samples are *independent*. This is not always the case!

Example

Suppose that 10 subjects have their blood pressure measured before and after taking a certain drug. For subject k, a *pair* of measurements (X_k, Y_k) is recorded, representing subject k's blood pressure before and after the medicine.

The variables X_k and Y_k are *not* independent, and in fact, the pair (X_k, Y_k) should be considered as a *single observation*.

We want an estimate (CI) of *difference* $\mu_X - \mu_Y$, where μ_X is the mean blood measure before taking the drug and μ_Y is the mean blood pressure after taking the drug.

Model and assumptions

- We begin with a random sample of *paired data*, i.e. we have a sample of pairs $(X_1, Y_1), (Y_2, Y_2), ..., (X_n, Y_n)$ such that the *pairs* are i.i.d. with the same *joint distribution*.
- Let μ_X and μ_Y denote the means of the marginal distributions for X_k and Y_k respectively. Likewise, let σ_X^2 and σ_Y^2 denote the variances for X_k and Y_k respectively.
- Furthermore, we assume that X_k and Y_k are **not** independent

Note:

$$\bar{X} - \bar{Y} = \frac{1}{n}\sum_{k=1}^{n} X_k - \frac{1}{n}\sum_{k=1}^{n} Y_k = \frac{1}{n}\sum_{k=1}^{n}(X_k - Y_k)$$

is an *unbiased* estimator of $\mu_X - \mu_Y$, but the variance of this estimator may be smaller than it is in the case of independent samples.

In particular, if $\mathbb{C}\text{ov}(\bar{X}, \bar{Y}) > 0$, then

$$\begin{aligned}
\mathbb{V}\text{ar}(\bar{X}_1 - \bar{X}_2) &= \mathbb{V}\text{ar}(\bar{X}) + \mathbb{V}\text{ar}(\bar{Y}) - 2\mathbb{C}\text{ov}(\bar{X}, \bar{Y}) \\
&= \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{n} - 2\mathbb{C}\text{ov}(\bar{X}, \bar{Y}) \\
&< \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{n} = \mathbb{V}\text{ar}(\bar{X}) + \mathbb{V}\text{ar}(\bar{Y}).
\end{aligned}$$

Confidence interval for $\mu_1 - \mu_2$ (paired data)

Let $D_k = X_k - Y_k$, then $\underline{D} = (D_1, D_2, ..., D_n)$ is a random sample and $\bar{D} = \bar{X}_1 - \bar{X}_2$ is an unbiased estimator for $\mu_1 - \mu_2$.

(1) If n is **large**, then

$$\frac{\bar{D} - (\mu_1 - \mu_2)}{\sigma_D / \sqrt{n}} \dot{\sim} \mathrm{N}(0, 1)$$

and $(\bar{D} - 1.96\frac{\sigma_D}{\sqrt{n}}, \bar{D} + 1.96\frac{\sigma_D}{\sqrt{n}})$ is an approximate 95% CI for $\mu_1 - \mu_2$.

If σ_D is *unknown*, it can be replaced by S_D.

(2) If n is small and the random sample $D_1, D_2, ..., D_n$ appear to be normally distributed, then

$$\frac{\bar{D} - (\mu_1 - \mu_2)}{S_D / \sqrt{n}} \sim t_{n-1}$$

and $(\bar{D} - t_{n-1,0.025}\frac{S_D}{\sqrt{n}}, \bar{D} + t_{n-1,0.025}\frac{S_D}{\sqrt{n}})$ is an (approximate) 95% CI for $\mu_1 - \mu_2$.

Blood Pressure Example cont.

The blood pressure of the 10 subjects measured before and after taking the drug is given by:

Subject	1	2	3	4	5	6	7	8	9	10
Before	156	159	145	160	151	143	164	157	149	160
After	155	148	146	147	140	141	148	150	149	154

Find a 95% CI for the true difference in the mean blood pressure before and after taking the drug, and use the interval to determine whether or not the drug is effective.

Confidence intervals for Bin(m, p)

Assume that we want to estimate a *proportion* of a population having a specific characteristic.

This can be expressed as a *probability* (e.g. probability next e-mail you receive contains a virus), *percentage* (e.g. percentage of people following a certain religion), or *rate* (e.g. mortality rate among pensioners of a certain age).

In many cases, we can express the above quantities as the probability p in a Bin(m, p) distribution.

Suppose that X is a *single* observation from Bin(m, p) (i.e., sample size is $n = 1$). The MLE for p is given by

$$\hat{p}(X) = \frac{X}{m}.$$

We would like to construct a confidence interval for p which is based on the MLE $\hat{p}$.

Confidence Intervals for Bin(m, p)

Goal: Estimate a population proportion having a specific characteristic.

This proportion can be expressed as:

- a *probability* (e.g., probability next email contains a virus),
- a *percentage* (e.g., percentage of people following a certain religion),
- a *rate* (e.g., mortality rate among pensioners of a certain age).

In many cases, these quantities correspond to the probability p in a Bin(m, p) distribution.

Suppose X is a single observation from Bin(m, p) (i.e., sample size $n = 1$). The maximum likelihood estimator (MLE) for p is:

$$\hat{p}(X) = \frac{X}{m}.$$

We want to construct a confidence interval for p based on the MLE $\hat{P}$.

Approximate CI for Bin(m, p)

To derive an approximate CI for p, recall that if $X \sim \text{Bin}(m, p)$ and m is **large**, the Central Limit Theorem gives

$$X \dot{\sim} \text{N}\left(mp, mp(1-p)\right)$$

This also implies that

$$\hat{p}(X) = \frac{X}{m} \dot{\sim} \text{N}\left(p, \frac{p(1-p)}{m}\right)$$

giving

$$\frac{\hat{p}(X) - p}{\sqrt{\frac{p(1-p)}{m}}} \dot{\sim} \text{N}\left(0, 1\right).$$

Now, as this expression involves the **unknown** true proportion p in a non-linear manner, we can further approximate it by using the estimate $\hat{p}$ to obtain an appropriate *pivotal quantity*

$$\frac{\hat{p}(X) - p}{\sqrt{\frac{\hat{p}(X)(1-\hat{p}(X))}{m}}} \dot{\sim} \text{N}\left(0, 1\right).$$

It follows that we can construct a $(1-\alpha)100\%$ CI for p, based on:

$$\mathbb{P}\left(-z_{\alpha/2} < \frac{\hat{p}(X)-p}{\sqrt{\frac{\hat{p}(X)(1-\hat{p}(X))}{m}}} < z_{\alpha/2}\right) \approx 1-\alpha$$

$$\mathbb{P}\left(\hat{p}(X) - z_{\alpha/2}\sqrt{\frac{\hat{p}(X)(1-\hat{p}(X))}{m}} < p < \hat{p}(X) + z_{\alpha/2}\sqrt{\frac{\hat{p}(X)(1-\hat{p}(X))}{m}}\right)$$
$$\approx 1-\alpha$$

The $(1-\alpha)100\%$ CI:

$$\left(\hat{p}(X) - z_{\alpha/2}\sqrt{\frac{\hat{p}(X)(1-\hat{p}(X))}{m}}, \quad \hat{p}(X) + z_{\alpha/2}\sqrt{\frac{\hat{p}(X)(1-\hat{p}(X))}{m}}\right)$$

CIs for differences between proportions

Suppose now that we want to estimate the difference between two proportions p_1 and p_2 based on two samples from the binomial populations $X_1 \sim \text{Bin}(m_1, p_1)$ and $X_2 \sim \text{Bin}(m_2, p_2)$.

For **large** m_1 and m_2 we can use the approximation

$$Z = \frac{(\hat{p}_1(X_1) - \hat{p}_2(X_1)) - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{m_1} + \frac{p_2(1-p_2)}{m_2}}} \dot{\sim} \text{N}(0, 1)$$

to obtain a $(1-\alpha)100\%$ CI for the difference $p_1 - p_2$ of the form

$$(\hat{p}_1(X_1) - \hat{p}_2(X_1)) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1(X_1)(1-\hat{p}_1(X_1))}{m_1} + \frac{\hat{p}_2(X_1)(1-\hat{p}_2(X_1))}{m_2}}.$$

Notice that, as before, the *unknown* proportions p_1 and p_2 have been substituted in the variance with their *estimates* $\hat{p}_1(X_1)$ and $\hat{p}_2(X_1)$, to give the above *approximate* CI.

Margin of Error

Suppose we survey $m = 1000$ people from a very large population, and $x = 300$ respond "yes" to a question. The goal is to estimate p, the proportion of people in the population who would answer "yes."

Assume:

$$X \sim \text{Bin}(1000, p), \quad \hat{p}(x) = \frac{x}{m} = 0.3.$$

A 95% confidence interval for p is:

$$\left(0.3 - 1.96\sqrt{\frac{(0.3)(0.7)}{1000}}, 0.3 + 1.96\sqrt{\frac{(0.3)(0.7)}{1000}}\right).$$

In other words, with 95% certainty:

$$|p - \hat{p}| \leq 1.96\sqrt{\frac{(0.3)(0.7)}{1000}} \approx 0.03.$$

So the **margin of error** is approximately 3%.

Exact CIs for Bin(m, p)

An exact 95% confidence interval for p can be obtained by finding values $p_L(X)$ and $p_U(X)$ such that:

$$\mathbb{P}\big(p_L(X) < p < p_U(X)\big) = 0.95.$$

Suppose $X = x$ is observed from Bin(m, p). Then:

$$p_L \text{ solves } \mathbb{P}(X \geq x \mid p = p_L) = \sum_{k=x}^{m} \binom{m}{k} (p_L)^k (1 - p_L)^{m-k} = 0.025,$$

and

$$p_U \text{ solves } \mathbb{P}(X \leq x \mid p = p_U) = \sum_{k=0}^{x} \binom{m}{k} (p_U)^k (1 - p_U)^{m-k} = 0.025.$$

These equations are solved numerically (using tables or software) to give an exact 95% CI for p of the form (p_L, p_U).

Example: Exact CI for p

Example: $m = 10$, $x = 4$.

Let $F(k; m, p)$ denote the CDF of $\text{Bin}(m, p)$, i.e.,

$$F(k; m, p) = \mathbb{P}(X \leq k), \quad X \sim \text{Bin}(m, p).$$

To find p_L and p_U, solve:

$$\mathbb{P}(X \geq 4 \mid p = p_L) = 1 - F(3; 10, p_L) = 0.025,$$

$$\mathbb{P}(X \leq 4 \mid p = p_U) = F(4; 10, p_U) = 0.025.$$

Using tables:

$$1 - F(3; 10, p_L) = 0.025 \Rightarrow F(3; 10, p_L) = 0.975 \Rightarrow p_L \approx 0.12,$$

$$F(4; 10, p_U) = 0.025 \Rightarrow 1 - F(5; 10, 1 - p_U) = 0.025 \Rightarrow 1 - p_U \approx 0.26$$

So the 95% CI for p is $(0.12, 0.74)$.

Recall: For $p > 0.5$, $F(k; m, p) = 1 - F(m - k - 1; m, 1 - p)$.